Peak Analysis Tool User Manual

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Introduction

The Peak Analysis Tool is a high-precision application for detecting, analyzing, and characterizing peaks in time-series data. It is designed for scientific use cases, such as single-molecule fluorescence or particle counting, where accurate peak detection and noise filtering are critical.

Getting Started

Installation

- **Windows**: Run `PeakService.exe`.
- **Source**: Run `python -m service.app`.

The application interface will open automatically in your default web browser at `http://localhost:8765`.

Workflow Overview

The application follows a structured workflow: 1. **Load**: Import raw data files. 2. **Process & Detect**: Filter noise and identify peaks. 3. **Analyze**: View time-series results and distributions. 4. **Export**: Save data and plots.

Loading Data

Navigate to the **Load Data** page to begin.

- **File Selection**: Click "Select Files" to choose `.txt`, `.xls`, or `.xlsx` files.
- **Protocol Information**: Enter metadata about your experiment (Date, Setup, Sample ID, Laser Power, etc.) to keep it associated with the dataset.
- **Settings**:
  - **Time Resolution**: Set the time per sample (e.g., 0.1 ms).
  - **Photon Correction**: Enable dead-time correction if using a photon counter. Enter the dead time in nanoseconds.

Recent sessions are saved and can be reloaded with a single click from the "Recent Sessions" list.

Process & Detect

This is the core analysis step.

1. Signal Filtering

Apply a digital filter to reduce noise before detection. - **Filter Type**: - **Butterworth**: General-purpose noise reduction. - **Savitzky-Golay**: Preserves peak shape better while smoothing. - **Parameters**: - **Cutoff Frequency (Hz)**: Frequencies above this are removed. - **Order**: Sharpness of the filter cutoff. - **Window Length**: Number of points for smoothing (Savitzky-Golay).

2. Peak Detection

Configure algorithms to find valid peaks. - **Prominence**: How much a peak stands out from the surrounding baseline. - **Min Distance**: Minimum samples between peaks. - **Width Min/Max**: Accepted width range in milliseconds.

- **Rel Height**: The relative height (0.0-1.0) at which width is measured. - **Prominence Ratio**: Filters noise by comparing prominence to absolute amplitude.

Click **Detect Peaks** to run the algorithm. Use **Auto Threshold** for an initial estimate.

Analysis & Export

View the detected peaks in context of the full time series.

- **Interactive Plot**: Zoom and pan to inspect individual peaks.
- **Throughput**: See the rate of peaks over time.
- **Histograms**: Analyze distributions of peak amplitudes and intervals.
- **Export**: Use the integrated Export controls to save peak data (CSV) or download the current chart view as an image.

Double Peak Analysis

For experiments involving paired events (e.g., double-labeled particles).

- **Pairing Logic**: Peaks are paired based on their temporal proximity.
- **Metrics**:
  - **Distance**: Time between paired peaks.
  - **Prominence Ratio**: Ratio of the second peak's intensity to the first.
  - **Width Ratio**: Comparison of peak shapes.
- **Thresholds**: Set min/max ranges for these metrics to filter valid pairs.

Exporting Results

Save your work for publication or further analysis.

- **Data Formats**: CSV, Excel, Text.
- **Metadata**: Option to include protocol details in the header.
- **Images**: Click the **Camera** icon on any chart to save a high-resolution PNG.

Preferences

Customize the application appearance (Light/Dark mode) and default analysis parameters.